

The Impact of Data Sparsity on Recommendation Algorithms: A Comparative Analysis of Memory-Based and Model-Based Approaches

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Abstract

Data sparsity remains a critical challenge for collaborative filtering algorithms, often compromising the delicate balance between recommendation accuracy and diversity. This study presents a comparative analysis of Memory-Based (kNN) and Model-Based (SVD) approaches using the MovieLens 1M dataset, employing a dual-track methodology of "Sparsity Injection" and "Density Filtering" to evaluate algorithmic robustness under stress. Our findings demonstrate that while kNN suffers linear degradation and generates stochastic noise under data loss, SVD maintains superior stability and organic exploration, proving effective for the cold start problem. Furthermore, we reveal a definitive trade-off in high-density environments where optimizing strictly for accuracy significantly reduces catalog coverage, effectively silencing the "Long Tail" of content. Consequently, this research identifies SVD as the superior architecture for sparse datasets and advocates for evaluation frameworks that treat accuracy and diversity as dual, equally critical objectives.

1. INTRODUCTION

In today's digital world, the biggest problem for users is not accessing information, it is finding the right content from the information overload. As **Prabakaran et al. (2023)** mentioned, the "Big Data" era is not just about the volume of data; it is also defined by its velocity and diversity. Giant platforms like Netflix or Amazon do not just store data. They process this data to understand user behavior and find products that users will be interested in. At this point, **Recommendation Systems** play a critical role. They analyze users' past preferences and predict their future interests.

The most common and effective method for these systems is the **Collaborative Filtering (CF)** technique, which is also the focus of our project. The logic of this method relies on a human concept called the "Homophily" (Similarity) principle in sociology. This means we assume that users who are similar to each other will like similar products in the future. In the literature, CF techniques are divided into two main categories. As **Aldayel et al. (2023)** described, these are **Memory-Based** methods and **Model-Based** methods. Memory-Based methods look directly at

the raw data in the database to make recommendations. Model-Based methods use machine learning techniques to build a prediction model from past data.

However, there is a big obstacle for these systems to work perfectly: **Data Sparsity**. In the real world, it is impossible for a user to rate all available movies or products. Most of the time, the user-item matrix is less than 1% full. As the data becomes sparse, traditional Memory-Based algorithms (like User-Based KNN) struggle to find similar users, and the system fails. At this point, the literature suggests **Model-Based** methods, such as **Matrix Factorization (SVD)**, as a strong alternative. These methods fill in the missing data with mathematical predictions and perform dimensionality reduction.

On the other hand, the success of a recommendation system is not just about "accuracy". If we constantly provide "safe" recommendations that the user already knows and likes, the system becomes monotonous. This creates the "**diversity**" problem. "Overspecialization" prevents the user from discovering new content. An ideal system should offer high accuracy but should not lose diversity (**Kunaver & Požrl, 2017**).

In this project, we will comparatively analyze the performance of Memory-Based (KNN) and Model-Based (Matrix Factorization) algorithms using the MovieLens dataset. The unique part of our work is testing the robustness of algorithms on artificially created sparsity levels (like removing 20%, 40%, 80% of data). Our goal is to observe how the balance between **accuracy (RMSE)** and **diversity (Diversity Score)** changes as data decreases. We aim to provide a numerical framework to decide which algorithm should be chosen at a specific sparsity threshold.

2. BACKGROUND AND RELATED WORK

Studies in the literature divide Collaborative Filtering techniques into two main groups based on their data approach: Memory-Based and Model-Based methods. **Aldayel et al. (2023)** explain this difference very clearly. Memory-Based systems connect directly to the database and use raw data to make recommendations. On the other hand, Model-Based systems use machine learning techniques to build a "model" from the data to make predictions.

2.1. Memory-Based Methods and the Sparsity Dead End

The traditional Memory-Based (User or Item-based) approach calculates the relationship between users using formulas like Cosine Similarity, as seen in the work of Murali et al.. However, the biggest enemy of these methods is the "Sparse Matrix" structure. In real life, for example on a movie platform, an average user rates only a very small number of movies (e.g., 50) out of tens of thousands. Even in the MovieLens 100k dataset, the Overall Sparsity (OS) ratio was observed to be as high as 0.937 (**Silva et al., 2018**).

This high sparsity is very bad for Memory-Based algorithms. Since users rate only a small portion of the movies, it creates the problem of "Reduced Common Ratings". Therefore, the reliability of metrics like the Pearson Correlation Coefficient drops significantly (**Silva et al., 2018**). Because the similarity scores are unreliable, the system cannot identify "neighbors with similar tastes." As a result, predictions are based on the wrong neighbors, and the error rate (RMSE) increases rapidly. In short, the recommendation mechanism stops working effectively.

2.2. Model-Based Solutions and Matrix Factorization

Model-Based approaches were developed to solve the failure of Memory-Based methods with sparse matrices. As **Dilip Yadav et al. (2024)** stated, the most effective solution family among these models is Matrix Factorization (MF) techniques. MF approaches rely on decomposing the sparse user-item matrix into low-rank matrices that represent "latent factors" for users and items (**Bojorque & Hurtado, 2025**). Thanks to their ability to learn these latent factors, these methods show better prediction accuracy on sparse data compared to traditional methods. This success supports our decision to choose **Singular Value Decomposition (SVD)**, a key member of MF, for our project.

2.3. Performance and Trade-Offs

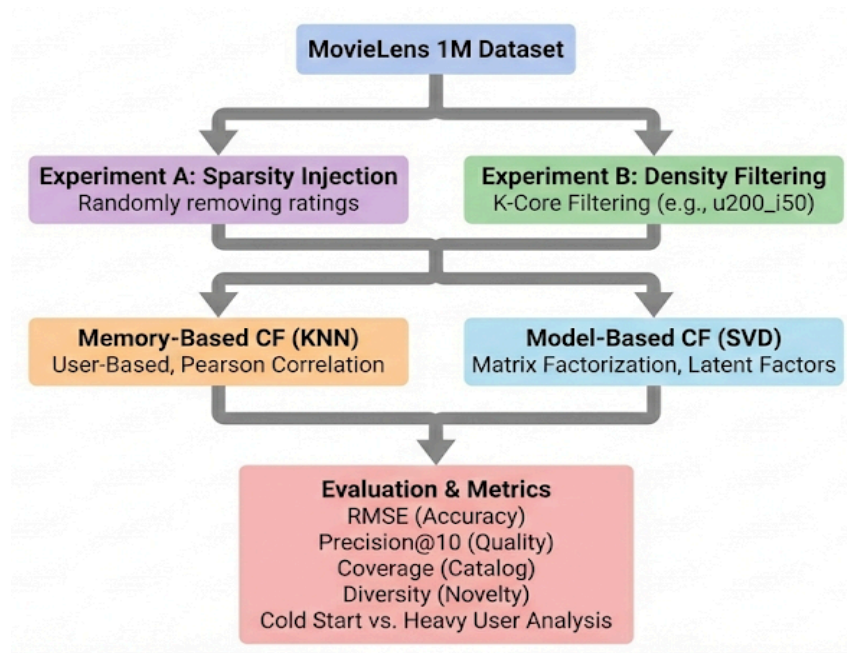
When comparing these two approaches, the analysis by **Aldayel et al. (2023)** shows a clear trade-off between accuracy and speed. The SAR algorithm, which uses Memory-Based logic, trains at an incredible speed of 0.83 seconds. However, it fails in accuracy and ranking success. On the other hand, Model-Based methods (like BiVAE and SVD) achieve much higher accuracy and better ranking quality (MAP/NDCG).

However, high accuracy in Model-Based methods does not always mean a "good recommendation." Studies on balancing accuracy and diversity generally fall into two categories: Two-stage strategies and Unified models. In two-stage systems, ratings are predicted first. Then, the list is re-ranked to increase diversity. However, this method cannot guarantee an optimal solution. In this context, the work by **Gogna and Majumdar (2017)** proposes a "single-stage unified optimization model" instead of simple re-ranking. This model aims to increase both "Individual Diversity" (ID) and "Aggregate Diversity" (AD) by looking at metadata like movie genres, not just ratings. Our project will also examine how much diversity strong models like SVD can provide by sacrificing some accuracy.

3. METHODOLOGY

(Project code URL: <https://github.com/fatmakaraca/DataCore>)

In this study, we followed a comparative approach to evaluate the robustness of recommendation algorithms. We designed a dual-track framework that tests algorithms under two opposite conditions: Sparsity Injection (removing data) and Density Filtering (cleaning noise). The overall workflow of our methodology is illustrated in Figure 1.



3.1. Algorithm 1: Memory-Based CF (KNN)

For the first approach, we used the User-Based K-Nearest Neighbors (KNN) algorithm. This method relies on the "Lazy Learning" principle. It does not build a model beforehand; instead, it searches for similar users directly in the database when a prediction is needed.

To calculate similarity between users, we chose the Pearson Correlation Coefficient instead of standard Cosine Similarity.

- Why Pearson? In real life, users have different rating habits. Some users are "generous" (always giving 5 stars), while others are "strict" (rarely giving high scores). This creates a "User Bias" problem. Pearson Correlation centers the data by subtracting the user's average rating. This normalizes the bias and provides a more accurate similarity score between users.

3.2. Algorithm 2: Model-Based CF (SVD)

For the second approach, we implemented Matrix Factorization using Singular Value Decomposition (SVD). Unlike KNN, this is a Model-Based method. It reduces the dimensionality of the massive user-item matrix to find hidden patterns.

- **Latent Factors:** SVD works by extracting "Latent Factors" (hidden features). For example, it can learn that a user likes "dark atmosphere" or "action scenes" without being explicitly told.
- **Optimization:** The model minimizes the error between the predicted rating and the real rating using Gradient Descent. This allows SVD to make predictions even for users with very little history, effectively handling sparse data.

4. RESULTS

4.1 EXPERIMENTAL SETUP

4.1.1. Tools and Environment: We implemented all algorithms using the Python programming language. The core of our recommendation logic relies on the Surprise Library, which is an industry-standard tool for Collaborative Filtering.

- **Data Manipulation:** We used Pandas and NumPy libraries to handle the MovieLens dataset and perform matrix operations.
- **Visualization:** We used Matplotlib and Seaborn to create graphs for RMSE, Diversity, and Coverage metrics.
- **Hardware:** The experiments were executed on the Kaggle Notebook environment. We utilized Kaggle's cloud-based computational resources (High-RAM CPUs) to process the large 1M dataset efficiently.

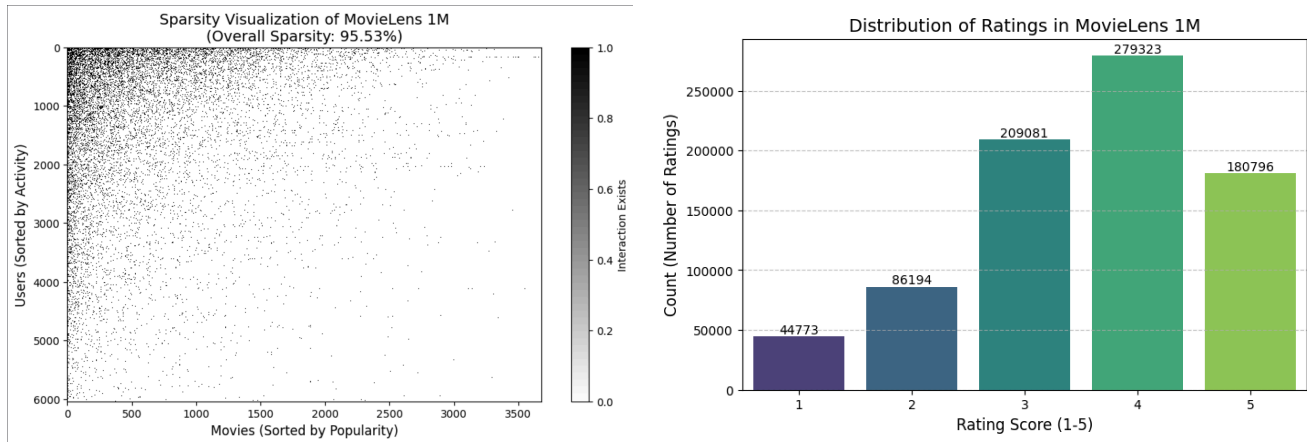
4.1.2. Model Tuning: To ensure fair comparison, we did not use random parameters. We applied Grid Search to find the optimal hyperparameters for the SVD model. We tuned critical parameters such as:

- **n_factors:** The number of latent factors (e.g., 50, 100).
- **lr_all:** The learning rate for the optimization steps.
- **reg_all:** The regularization term to prevent overfitting. For KNN, we optimized the number of neighbors (**k**) to balance accuracy and speed.

4.3.1. Dataset

In this study, the MovieLens 1M dataset, which is an industry standard in recommendation systems, was used. This dataset, collected by the GroupLens Research Laboratory, contains user ratings for movies on a scale of 1 to 5. It includes 6040 users, 3,706 movies, and approximately 1000209 ratings, and the original version of our dataset has a sparsity level of 95.53%.

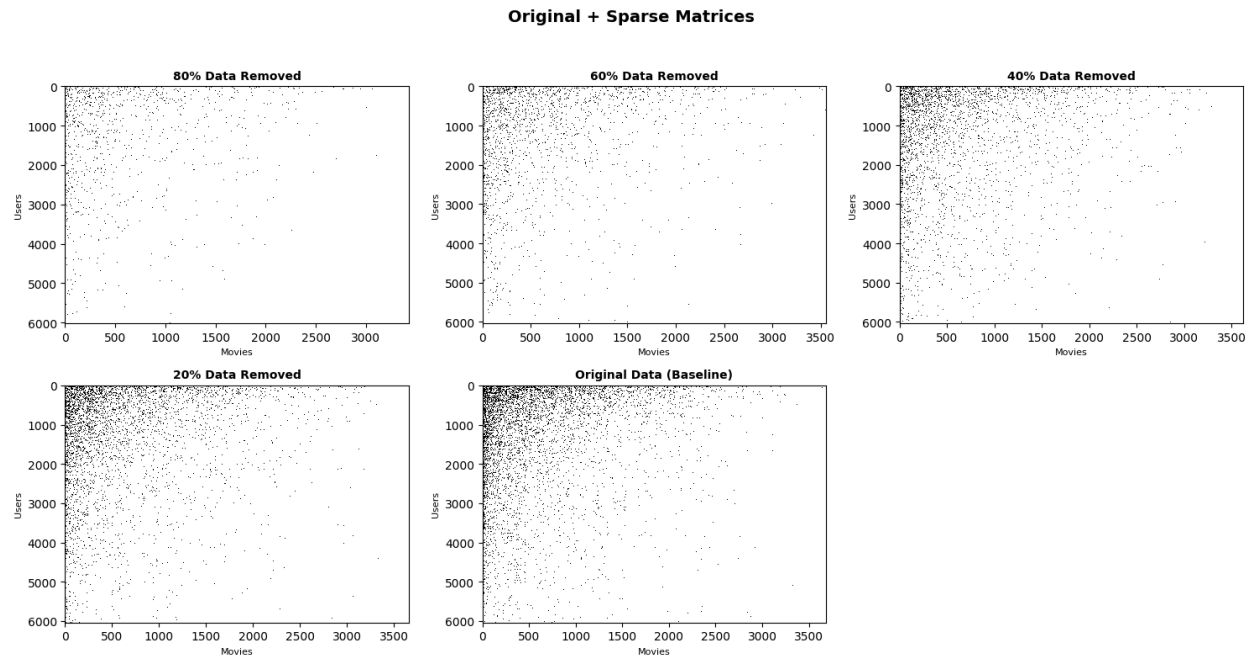
The total capacity of the matrix, calculated by multiplying 6040 users by 3706 movies, corresponds to approximately 22.3 million possible interactions. However, the number of filled cells in the dataset is around 1 million. With a sparsity of 95% and a density level of 4.47%, this dataset is highly suitable for the purpose of our study, as it allows us to compare the performance of two different approaches on sparse matrices. Moreover, this high level of sparsity directly simulates the "information scarcity" problem faced by popular modern platforms that use recommendation systems, such as Netflix and Amazon. The MovieLens 1M dataset we selected provides an excellent basis for testing how algorithms handle sparse matrices.



4.2 EXPERIMENT 1: THE IMPACT OF SPARSITY INJECTION

In this experiment, the aim was to evaluate the resilience of the Memory-Based approach (KNN) and the Model-Based approach (SVD) against data sparsity. According to our hypothesis, the performance of the KNN algorithm is expected to degrade as the data becomes sparser, whereas SVD is anticipated to maintain its model performance with considerable robustness.

To test the algorithms under different levels of sparsity, a stress test called "Sparsity Injection" was applied. Using our original dataset, four different sparsity levels—20%, 40%, 60%, and 80%—were generated according to the MCAR (Missing



Completely at Random) principle.

The experimental results obtained revealed a dramatic difference in behavior between the two algorithm families:

- **Memory-Based (KNN):** The User-Based KNN algorithm experienced a significant performance drop as the data became sparser. In particular, at 80% sparsity, the RMSE value increased substantially. KNN operates on the principle of overlapping data. As data points were removed, the number of common movies between users decreased, causing similarity metrics such as the Pearson Correlation to lose reliability. As a result, the model was unable to find enough neighbors to make accurate predictions.
- **Model-Based (SVD):** Thanks to the "Matrix Factorization" technique, the SVD algorithm generalizes the data instead of memorizing it, treating missing values not as "gaps" but as values to be predicted through the product of latent feature vectors associated with users and movies. As a result, SVD achieved higher performance than KNN on the original data, even at an 80% sparsity level. Even when users had no movies in common, it was able to make accurate predictions thanks to the learned latent features.

Sparsity Level	Algorithm	RMSE	Diversity	Coverage
Original	KNN	0.9651	0.7903	67.24%
Original	SVD	0.8763	0.7690	62.84%
20%	KNN	0.9713	0.7916	67.45%
20%	SVD	0.8788	0.7963	62.61%
40%	KNN	0.9790	0.7936	70.10%
40%	SVD	0.8859	0.8466	62.37%
60%	KNN	0.9903	0.8034	79.73%
60%	SVD	0.9009	0.9144	62.45%
80%	KNN	1.0401	0.8188	88.15%
80%	SVD	0.9305	0.9818	63.17%

As presented in Table, the numerical data supports these behavioral observations. While SVD maintained a notably lower error rate (RMSE 0.9305) even at extreme sparsity (80%), KNN's error escalated to 1.0401. A striking finding was observed in the Coverage metric. Paradoxically, KNN's coverage increased from 67.24% to 88.15% as data vanished. In this context, this increase represents "noise" rather than high-quality exploration. As KNN lost true neighbors due to sparsity, it began selecting random neighbors, leading to scattered and irrelevant recommendations across the catalog. In contrast, SVD maintained a stable coverage (~62%) while significantly improving its Diversity score (from 0.76 to 0.98), proving its ability to find niche items through latent features without succumbing to random noise.

4.3 EXPERIMENT 2: THE EFFECT OF DENSITY FILTERING:

While the first experiment focuses on algorithmic robustness under stress, this second experiment focuses on the **"performance ceiling"** of the models by minimizing noise. To obtain this desired information, iterative K-core filtering was applied to progressively increase the density of the data. The filtering levels ranged from "original data's density" to a higher density, denoted as u200_i50, which means only users with at least 200 ratings and items with at least 50 ratings were retained. This approach increased the matrix density to approximately 17%.

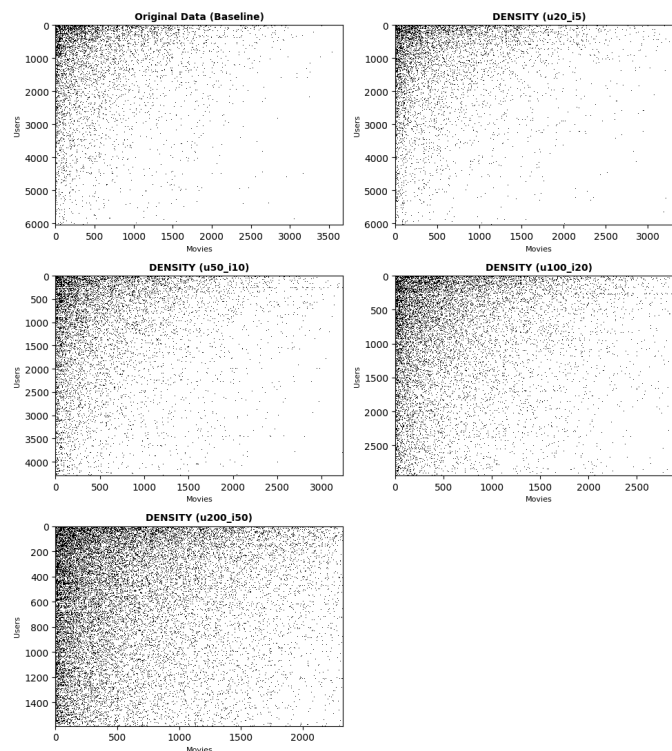
The results demonstrated a clear inverse relationship between data density and error rates. As the dataset density increased, the RMSE for both algorithms exhibited a marked decline, directly

translating to enhanced predictive accuracy. The SVD model achieved its optimal performance at the highest density level, recording an RMSE of 0.85. These findings corroborate the hypothesis that Matrix Factorization techniques can learn highly precise user and item vectors when provided with rich interaction histories.

However, this accuracy gain revealed a critical trade-off regarding the “Long Tail”. As illustrated in our coverage analysis, filtering for density caused a decline in catalogue coverage for both models, reaching a low of approximately 20%. This indicates that while the system achieved high accuracy for popular items, it effectively lost the ability to recommend various content. We observed that SVD’s diversity scores decreased in this dense environment (from 0.77 to 0.60), suggesting that with high-quality data, the model converges on a specific set of "best-fit" popular items rather than exploring the catalog. Furthermore, while Matrix Factorisation inherently helps mitigate the 'filter bubble' effect, it is crucial to acknowledge that excessive density filtering can still suppress coverage to a point where the model’s overall diversity is severely compromised.

Consequently, a primary finding of this experiment is that relying on accuracy metrics, such as RMSE, doesn’t provide a complete assessment of system performance. Therefore, this study suggests that a robust evaluation framework must treat accuracy and diversity as dual objectives. An optimal recommender system should not be judged only by its ability to minimize error, but by its capacity to balance high precision with sufficient catalog exposure.

Dense Matrices + Original



5. CONCLUSION

This study conducted a comparative analysis of Memory-Based (kNN) and Model-Based (SVD) collaborative filtering, revealing a fundamental trade-off between robustness and exploration. We applied sparsification and densification to the dataset, respectively, to derive conclusions about how data sparsity affects recommendation quality.

The first finding of this study is that, as our stress tests show, the Model-Based approach is the superior choice. While kNN suffers from linear degradation due to the loss of neighbor correlations, SVD's latent factor approach remains stable even when 80% of the data is missing, proving its effectiveness in handling the "Cold Start" problem.

Second, our behavioral analysis highlighted the distinction between noise and discovery. We found that the high diversity scores observed in kNN under high sparsity were artifacts of stochastic noise rather than true exploration. However, SVD provided an "organic exploration," maintaining stable catalog coverage regardless of data sparsity.

Finally, we explored an Accuracy-Coverage Trade-off. In dense data environments, both models achieved their peak accuracy, but at the high cost of catalog coverage. This confirms that optimizing strictly for RMSE often biases the system toward popular items, ignoring the "Long Tail" of the catalog. This contributes to the literature by highlighting that evaluation frameworks optimize strictly for RMSE risk bias systems toward popular items, effectively silencing niche content.

Building on these findings, future research should address the limitations of pure Collaborative Filtering by exploring "hybrid" recommender systems. By integrating the structural robustness of SVD with content-based features, such as movie genres, directors, or user demographics, the system could further mitigate the "Cold Start" problem by utilizing metadata when interaction history is absent. Furthermore, given the trade-off between accuracy and catalog coverage, future studies should investigate "Multi-Objective Optimization" frameworks or re-ranking algorithms. These methods would enable a systematic balance between RMSE and diversity metrics. Finally, the potential of Deep Learning architectures should be explored to assess whether non-linear feature interactions can surpass linear Matrix Factorization in highly sparse environments.

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